

Red 1 $-1 \leq x \leq 1$

$$\theta_{10} = 0 \quad \theta_{11} = -1 \quad \rightarrow \quad \theta_{10} + \theta_{11}x = -x$$

$$\theta_{20} = 0 \quad \theta_{21} = 1 \quad \rightarrow \quad \theta_{20} + \theta_{21}x = x$$

$$\theta_{30} = -0.67 \quad \theta_{31} = 1 \quad \rightarrow \quad \theta_{30} + \theta_{31}x = x - 0.67$$

$$\phi_0 = 1 \quad \phi_1 = -2 \quad \phi_2 = -3 \quad \phi_3 = 9.3$$

$$a = \text{ReLU}$$

$$h_1 = a[\theta_{10} + \theta_{11}x] = \text{ReLU}(-x) = \begin{cases} -x & -1 \leq x < 0 \\ 0 & 0 \leq x \leq 1 \end{cases}$$

$$h_2 = a[\theta_{20} + \theta_{21}x] = \text{ReLU}(x) = \begin{cases} 0 & -1 \leq x < 0 \\ x & 0 \leq x \leq 1 \end{cases}$$

$$h_3 = a[\theta_{30} + \theta_{31}x] = \text{ReLU}(x - 0.67) = \begin{cases} 0 & -1 \leq x < 0.67 \\ x - 0.67 & 0.67 \leq x \leq 1 \end{cases}$$

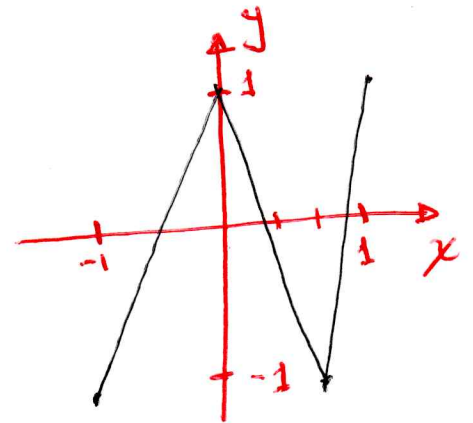
$$\phi_1 h_1 = \begin{cases} 2x & -1 \leq x < 0 \\ 0 & 0 \leq x \leq 1 \end{cases}$$

$$\phi_2 h_2 = \begin{cases} 0 & -1 \leq x < 0 \\ -3x & 0 \leq x \leq 1 \end{cases}$$

$$\phi_3 h_3 = \begin{cases} 0 & -1 \leq x < 0.67 \\ 9.3x - 6.231 & 0.67 \leq x \leq 1 \end{cases}$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$

$$y = \begin{cases} 2x + 1 & -1 \leq x < 0 \\ -3x + 1 & 0 \leq x < 0.67 \\ 6.3x - 5.231 & 0.67 \leq x \leq 1 \end{cases}$$



Red 2

$$-1 \leq y \leq 1$$

$$\theta'_{10} = -0.6 \quad \theta'_{11} = -1 \rightarrow \theta'_{10} + \theta'_{11}y = -y - 0.6$$

$$\theta'_{20} = 0.2 \quad \theta'_{21} = 1 \rightarrow \theta'_{20} + \theta'_{21}y = y + 0.2$$

$$\theta'_{30} = -0.5 \quad \theta'_{31} = 1 \rightarrow \theta'_{30} + \theta'_{31}y = y - 0.5$$

$$h_1' = a[\theta'_{10} + \theta'_{11}y] = \text{ReLU}(-y - 0.6) = \begin{cases} -y - 0.6 & -1 \leq y < -0.6 \\ 0 & -0.6 \leq y \leq 1 \end{cases}$$

$$h_2' = a[\theta'_{20} + \theta'_{21}y] = \text{ReLU}(y + 0.2) = \begin{cases} 0 & -1 \leq y < -0.2 \\ y + 0.2 & -0.2 \leq y \leq 1 \end{cases}$$

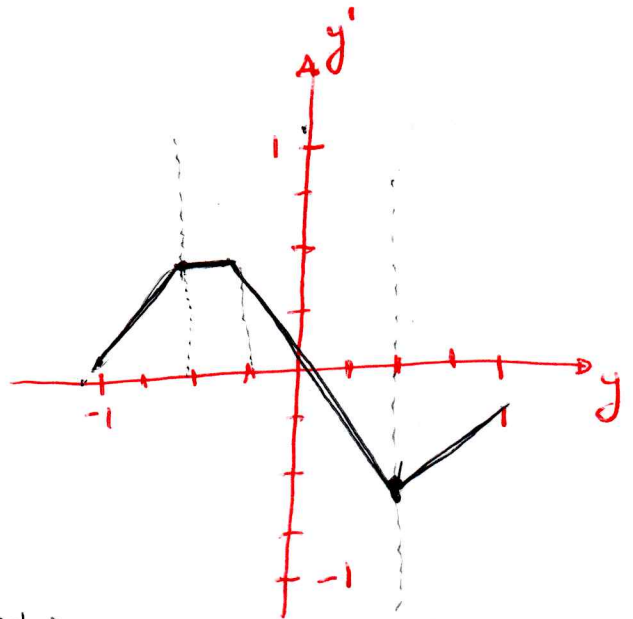
$$h_3' = a[\theta'_{30} + \theta'_{31}y] = \text{ReLU}(y - 0.5) = \begin{cases} 0 & -1 \leq y < 0.5 \\ y - 0.5 & 0.5 \leq y \leq 1 \end{cases}$$

$$\phi'_0 = 0.5 \quad \phi'_1 = -1 \quad \phi'_2 = -1.5 \quad \phi'_3 = 2$$

$$\phi'_1 h_1' = \begin{cases} y + 0.6 & -1 \leq y < -0.6 \\ 0 & -0.6 \leq y \leq 1 \end{cases}$$

$$\phi'_2 h_2' = \begin{cases} 0 & -1 \leq y < -0.2 \\ -1.5y - 0.3 & -0.2 \leq y \leq 1 \end{cases}$$

$$\phi'_3 h_3' = \begin{cases} 0 & -1 \leq y < 0.5 \\ 2y - 1 & 0.5 \leq y \leq 1 \end{cases}$$



$$y' = \phi'_0 + \phi'_1 h_1' + \phi'_2 h_2' + \phi'_3 h_3'$$

$$y' = \begin{cases} y + 1.1 & -1 \leq y < -0.6 \\ 0.5 & -0.6 \leq y < -0.2 \\ -1.5y + 0.2 & -0.2 \leq y < 0.5 \\ 0.5y - 0.8 & 0.5 \leq y \leq 1 \end{cases}$$

$$-1 \leq y \leq 1$$

$$y = 2x + 1$$

$$y = -3x + 1$$

$$y = 6.3x - 5.231$$

$$y = -1$$

$$x = -1$$

$$x = \frac{2}{3} \approx 0.67$$

$$x = \frac{4.231}{6.3} \approx 0.67$$

$$y = -0.6$$

$$x = \frac{-1.6}{2} = -0.8$$

$$x = \frac{1.6}{3} \approx 0.53$$

$$x = \frac{4.631}{6.3} \approx 0.74$$

$$y = -0.2$$

$$x = \frac{-1.2}{2} = -0.6$$

$$x = \frac{1.2}{3} = 0.4$$

$$x = \frac{5.031}{6.3} \approx 0.8$$

$$y = 0.5$$

$$x = \frac{-0.5}{2} = -0.25$$

$$x = \frac{1}{6} \approx 0.17$$

$$x = \frac{5.731}{6.3} \approx 0.91$$

$$y = 1$$

$$x = 0$$

$$x = 0$$

$$x = \frac{6.231}{6.3} \approx 0.99$$

$$y =$$

$$\underline{2x + 2.1}$$

$$\underline{0.5}$$

$$-1.5(2x + 1) + 0.2$$

$$\underline{-3x - 1.3}$$

$$0.5(2x + 1) - 0.8 = \underline{x - 0.3}$$

$$0.5(-3x + 1) - 0.8$$

$$\underline{-1.5x - 0.3}$$

$$-1.5(-3x + 1) + 0.2 = \underline{4.5x - 1.3}$$

$$\underline{0.5}$$

$$\underline{-3x + 2.1}$$

$$\underline{6.3x - 4.131}$$

$$\underline{0.5}$$

$$-1.5(6.3x - 5.231) + 0.2$$

$$\underline{-9.45x + 8.0465}$$

$$0.5(6.3x - 5.231) - 0.8$$

$$\underline{3.15x - 3.4155}$$

$$-1 \leq x < -0.8$$

$$-0.8 \leq x < -0.6$$

$$-0.6 \leq x < -0.25$$

$$-0.25 \leq x < 0$$

$$0 \leq x < 0.17$$

$$0.17 \leq x < 0.4$$

$$0.4 \leq x < 0.53$$

$$0.53 \leq x < 0.67$$

$$0.67 \leq x < 0.74$$

$$0.74 \leq x < 0.8$$

$$0.8 \leq x < 0.91$$

$$0.91 \leq x \leq 0.99$$

